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Redefining volatility forecasting in the aerospace and defense sector: application of CEEMDAN-GARCH models

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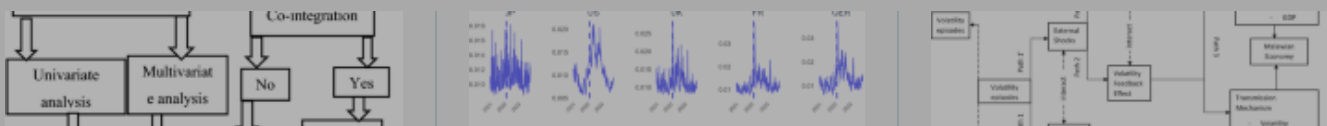
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Results indicate that CEEMDAN offered modest improvements in forecasting accuracy for selected indices and models, without fundamentally altering the best-performing model rankings. This exploratory study contributes to the volatility forecasting literature by demonstrating the contextual applicability of a CEEMDAN-GARCH hybrid framework for complex financial time series. The findings offer preliminary insights into risk management and decision-making for investors, policymakers, and industry participants in a rapidly evolving security landscape.

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economy and financial markets, manifesting through various channels, including the imposition of economic sanctions, fluctuations in commodity prices, and disruptions in global supply chains (Alsayed, [2023](#)). Consequently, global stock markets have witnessed significant volatility, albeit with variations across countries depending on their economic circumstances and geopolitical relationships, particularly those with direct proximity or political ties to Ukraine and Russia (Boungou and Yatié, [2022](#)). The aerospace and defense sector is of no exception, as it has recently gained momentum driven by heightened security concerns stemming from the conflict between Russia and Ukraine.

Actually, for years, investors have steered clear of investing in aerospace and defense stocks to comply with ESG (environmental, social, and governance) mandates. This avoidance stems from the involvement of aerospace and defense firms in manufacturing

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The immediate necessity for investors and investment firms is therefore to study the dynamics of aerospace and defense companies in terms of returns and volatility to better inform their decisions. While previous studies have examined volatility forecasting for general financial indices, they often neglect sector-specific indices such as aerospace and defense, especially those outside the United States (McDonald and Kendall, [1994](#); Bae and Duvall, [1996](#); Alaminos et al., [2024](#)). This paper seeks to address this gap by analyzing the volatility of aerospace and defense indices from four countries—Korea, China, France, and the United Kingdom. The selection of China, South Korea, France, and the United Kingdom is based on several key criteria. These countries are major players in the global aerospace and defense sector, each with significant contributions to military expenditures and technological advancements. Their inclusion ensures geopolitical and economic diversity, capturing the varied effects of global disruptions such as the COVID-19 pandemic and the Russia-Ukraine conflict.

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asymmetric volatility shocks, which traditional econometric models may struggle to capture effectively. Additionally, the sector faces structural changes due to the interplay between ESG pressures—which discourage investment in defense stocks—and a renewed focus on national security, leading to fluctuating investor sentiment. These unique dynamics contribute to nonstationary, nonlinear, and clustered volatility patterns, necessitating a decomposition-based approach like CEEMDAN to extract meaningful trends from complex financial signals. Given that volatility studies in aerospace and defense remain underexplored, this research extends the application of CEEMDAN to a sector where standard GARCH models alone may not fully capture the impact of geopolitical risks and policy-driven volatility.

The complexity of financial markets, driven by their inherent non-stationarity, clustering, and asymmetric volatility, necessitates sophisticated analytical tools capable of

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Standard Generalized Autoregressive Conditional Heteroskedasticity model (SGARCH), and Glosten-Jagannathan-Runkle Generalized Autoregressive Conditional Heteroskedasticity (GJR GARCH) which are designed to capture persistence, asymmetry, and structural breaks in financial time series (Ghamgosar and Alizadeh, [2023](#)). By applying this hybrid CEEMDAN-GARCH approach to aerospace and defense indices, the study not only overcomes the limitations of standalone GARCH models but also contributes a novel methodological framework to the volatility literature. This is particularly relevant as the aerospace and defense sector is characterized by complex supply chains, cyclicalities, and sensitivity to political and defense spending shifts (Ivan and Lucia, [2013](#)).

Moreover, this study introduces a comparative analysis framework that examines the volatility of aerospace and defense indices across four diverse geopolitical and

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Which GARCH-type model (IGARCH, SGARCH, or GJR-GARCH) performs optimally in forecasting volatility for aerospace and defense indices in China, South Korea, France, and the United Kingdom?

3. What are the sector-specific volatility dynamics in response to global crises such as the COVID-19 pandemic and the Russia-Ukraine conflict?

The contributions of this paper are multifaceted. First, it fills a significant gap in the literature by focusing on aerospace and defense indices—an area previously underexplored in volatility studies. Second, it establishes the value of combining CEEMDAN with GARCH-type models to capture and predict the unique multiscale dynamics of these indices, particularly in response to global events. Third, it provides an actionable, cross-national analysis that informs risk management and decision-making

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discussion of the results. Finally, Section “Conclusion” offers concluding remarks and provides recommendations.

Literature review

Volatility forecasting remains a critical area of research in financial economics, driven by the need to understand and predict market behavior amidst increasing complexity and external disruptions. Traditional econometric models, particularly GARCH and its variants, have been extensively utilized for this purpose, given their ability to model conditional volatility. These models assume stationarity in financial time series, making them effective for relatively stable datasets. However, their limitations become evident when applied to high-frequency, non-stationary, or nonlinear data, such as indices in highly dynamic sectors like aerospace and defense. Researchers, including Li et al.

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approach allowed for a more robust extraction of IMFs, enabling analysts to capture complex patterns in non-stationary and nonlinear data with greater reliability (AL-Musaylh et al., [2018](#); Celik et al., [2020](#)). The introduction of white noise essentially spread the energy of different signal components across a wider spectrum, reducing the likelihood of mode mixing and improving the overall clarity of the decomposition process.

However, despite its theoretical and practical advancements, EEMD introduced a new challenge: reconstruction errors. The addition of white noise, while beneficial for separating modes, often distorted the original signal. This distortion could compromise the accuracy of the reconstructed time series, particularly when the magnitude or distribution of the added noise was not carefully calibrated (Zhang et al., [2021](#)). Such reconstruction errors limited the efficacy of EEMD in applications requiring high

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rendered CEEMD less effective when dealing with signals that exhibited significant fluctuations in their local energy or volatility. In such cases, the static noise addition could either fail to adequately separate the modes or overcompensate, introducing unnecessary complexity into the IMFs. These limitations highlighted the need for further refinement in decomposition techniques, ultimately paving the way for the development of Complementary Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN). CEEMDAN addressed the shortcomings of both EEMD and CEEMD by dynamically adjusting the noise amplitude based on the local energy of the signal. This adaptation ensured that the decomposition process remained robust across a wide range of signal properties, setting a new standard for accuracy and versatility in signal processing (Torres et al., [2011](#)).

Studies such as those by Cao et al. ([2019](#)), Han et al. ([2024](#)), Huang et al. ([2020](#)), Lin et

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predicting exchange rates and futures prices. Zhang et al. ([2015](#)) demonstrated that the integration of EEMD with GARCH models significantly improved the forecasting accuracy of commodity prices, particularly in scenarios characterized by extreme volatility. Cheng and Wei ([2014](#)) introduced a hybrid time-series support vector regression (SVR) model based on EMD for predicting stock prices of the Taiwan Stock Exchange Capitalization Weighted Stock Index (TAIEX). Nava et al. ([2018](#)) introduced a simple approach that combines EMD and Pearson's cross-correlations over rolling windows to quantify dynamic dependency at different time scales for three financial indices. Zhang et al. ([2020](#)) used a model that integrates CEEMD and machine learning techniques to improve the predictive accuracy of stock indices. Aggarwal et al. ([2020](#)) compared the original Bitcoin price series with decomposed sequences from EEMD and CEEMDAN. The error for CEEMDAN was nearly zero (max value: 1.6×10^{-11}), while EEMD showed significantly larger errors (max value: 132.76), proving CEEMDAN's superior ability to

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preference for these sectors due to their expanding role in sustainability. Similarly, Zhang et al. ([2022](#)) investigated co-movements between the Geopolitical Risk (GPR) index and returns from 36 global defense and aerospace companies using wavelet coherence, highlighting the sector's sensitivity to geopolitical uncertainties. The results reveal significant co-movement at medium and longer scales around the onset of the war in Ukraine, indicating a flight-to-arms phenomenon. This effect is notably strong among US and European companies, where the GPR index positively influences returns and volatility during the war period. Bouri et al. ([2024](#)) applied a quantile-based model to examine return and volatility spillovers among 21 global aerospace companies, particularly during periods of heightened geopolitical tensions such as the COVID-19 pandemic and the Russia-Ukraine conflict. The findings reveal that return and volatility spillovers fluctuate over time and intensify during the COVID-19 and Russia-Ukraine war periods, particularly at extreme quantiles, confirming that geopolitical risk significantly

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on the abnormal returns of five major U.S. defense companies' stock prices over a 30-year period (1990–2019). Using a fixed effect panel model and event study methodology, the results show that defense budget announcements have a complex and dynamic positive impact on stock performance, and direct military conflict participation leads to a positive impact on stock performance, although this is moderated by a negative reaction in subsequent periods. (Gheorghe and Panazan, [2024](#)) examined the influence of geopolitical risk on the stock returns of 75 global representative defense companies from 2014 to 2023. Using wavelet coherence and phase differences to examine daily datasets, the results reveal that the Crimean annexation, the COVID-19 pandemic, and the Russia-Ukraine war significantly affected these companies, with European, U.S., and South Korean firms reacting notably to geopolitical events.

These studies provide valuable insights into how geopolitical shocks influence the

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investor sentiment in this industry is uniquely polarized, as ESG-related concerns previously led to divestment from defense stocks, whereas recent geopolitical crises have driven renewed capital inflows, creating nonlinear, asymmetric market reactions. Given these complexities, applying CEEMDAN within a GARCH framework enables a more refined analysis of multi-scale volatility drivers, distinguishing between short-term shocks (e.g., war escalations) and long-term trends (e.g., sustained defense spending shifts). This sector-specific behavior further justifies the need for decomposition techniques that can isolate structural volatility patterns more effectively than traditional financial modeling approaches.

In this study, standard GARCH models serve as a baseline for comparison to assess the effectiveness of CEEMDAN-enhanced volatility forecasting. Specifically, we apply three widely used GARCH-type models—SGARCH, IGARCH, and GJR-GARCH—to model the

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The EMD algorithm

Let $x(t)$ be a signal dependent on time t . Assume that $x(t)$ consists of a pure oscillatory component, $c(t)$, characterized by proper rotation with a mean of zero, and a residual term, $r(t)$, which represents the trend or baseline signal after removing the oscillatory component. EMD, as originally proposed, is implemented using a sifting process (steps (1) to (4)), summarized as follows:

(1) Locate the successive extrema of $x(t)$. Then, connect the local maxima with a natural cubic spline to form the upper envelope, $\{e_{\max}^{(1,1)}(t)\}$; and similarly connect the local minima to form the lower envelope, $\{e_{\min}^{(1,1)}(t)\}$.

(2) Calculate the mean $\{m^{(1,1)}(t)\}$ of both envelopes as a function of time.

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$$h_{1,2}(t) = h_{1,1}(t) - m_{1,2}(t)$$

Evaluate whether $h_{1,2}(t)$ fulfills the IMF criteria or the stoppage criterion. If it does, it is designated as the first IMF component. If not, continue iterating steps 1 to 3 with the updated proto-IMF candidates $h_{1,2}(t)$, $h_{1,3}(t) = h_{1,2}(t) - m_{1,3}(t)$ until a proto-IMF satisfies the required conditions to be classified as an IMF or meets the stoppage criterion.

This first IMF will be denoted $H_1(t)$, representing the component of the original signal with the finest amplitude (or in other terms, the shortest period).

(4) Compute the residuals

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Notably, algebraic manipulations in the above process reveal that:

$$x(t) = \sum_{k=1}^K H_k(t) + R_K(t)$$

(5)

Additionally, all IMF components, except the first, can be expressed as sums of spline functions. As a result, the IMFs can accurately reconstruct the original signal.

According to Wang et al. ([2010](#)), the definition of an IMF (as presented earlier) is not the only crucial element in EMD. A stoppage criterion is typically required to determine the number of iterations necessary to obtain a new IMF, as rigorously verifying the conditions for a proto-IMF to qualify as a true IMF is challenging in practice.

Consequently, selecting an appropriate stoppage criterion becomes a key element in

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$$SD_{ij}(t)=m_{k,j}(t)$$

(7)

Where $\{m\}_{k,j}(t)$ is the mean of the envelopes as a function of time, (k) is the number of IMFs, and (j) represents the iterations in the sifting process.

Requiring that the mean of the envelopes be smaller than the threshold τ , implying symmetry of the envelopes. (ii) The S-number criterion (Huang et al., [2003](#)), which is based on the number of consecutive siftings for which the number of zero-crossings and extrema are equal or at most differing by one (the S-number). An S-number is pre-specified, and the sifting process will stop only if, for S consecutive siftings, the numbers of zero-crossings and extrema stay the same and are equal or at most differ by one. The

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Ensemble Empirical Mode Decomposition (EEMD)

EEMD proposed by Wu and Huang ([2011](#)), is a noise-assisted strategy designed to alleviate (and in some cases, overcome) the mode mixing problem by adding a Gaussian white noise, which has a uniform time-frequency distribution, to the original signal.

$$x^{(i)}(t) = x(t) + \pi^{(i)}(t), i = 1, \dots, m$$

(8)

Where $x(t)$ represents the original signal, and $\pi^{(i)}(t)$ is the i^{th} realization of white noise added to the signal.

Specifically, the core idea of EEMD is to apply the EMD algorithm over multiple

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application of the EMD algorithm. One issue is that the reconstructed signal, obtained by summing the modes, may include noise that was not present in the original signal. Another challenge is that different realizations of the EMD decomposition over varied incremented signals $\{x^{*i}(t)\}$ may result in a different number of IMFs, potentially compromising the accuracy of the averaging process.

Complete EEMD (CEEMD)

Since the white noise added during the EEMD procedure to address the mode mixing problem might not be completely eliminated during the decomposition process (especially when the ensemble size m is small), the interaction between the signal and the noise can result in the generation different modes (Rahimpour et al., [2021](#)). The CEEMD algorithm, however, eliminates completely the residual white noise by adding

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Decompose both sets of modified original signals using the EMD algorithm to generate two sets of ensemble IMFs. The ensemble IMFs obtained from the positive mixtures contribute to one set of IMFs with positive residuals of the added white noise. Similarly, the ensemble IMFs obtained from the negative mixtures contribute to another set of ensemble IMFs with negative residuals of the added white noise.

Obtain the K^{th} IMF, $\{H_{K^i}(t)\}_{i=1,\dots,m}$, by averaging the corresponding ensemble of both IMFs with positive and negative noise. Continue this process to find the K IMFs. In contrast to EEMD, CEEMD ensures that the IMF set is quasi-complete and orthogonal. However, CEEMD is computationally intensive and may require a substantial amount of time to execute.

Complete EEMD with adaptive noise (CEEMDAN)

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(1) The algorithm begins by adding a Gaussian white noise $\{\pi^i\}_{i=1}^m(t)$ with a standard normal distribution to the original signal $x(t)$:

$$\begin{array}{c} x^i(t) = x(t) + \pi^i(t) \\ \text{with } i=1,2,\dots,m \end{array}$$

(12)

This noise addition is performed m times to create an ensemble of noisy signals, preserving the statistical characteristics of the original signal.

(2) Each noisy signal $\{x^i(t)\}$ is decomposed using EMD to extract the first IMF, denoted as $\{IMF_1^i(t)\}$.

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This step is repeated m times. Each noisy residual is then decomposed using EMD to extract the next order component:

$$\mathcal{IMF}_2(t) = \frac{1}{m} \mathop{\sum} \limits_{i=1}^m \mathcal{IMF}_1^i(t) \quad (15)$$

The corresponding residual is computed as:

$$R_2(t) = R_1(t) - \mathcal{IMF}_2(t) \quad (16)$$

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property assessments of the IMFs, are conducted. Additionally, the forecasting accuracy of the model is evaluated using out-of-sample data, ensuring that the CEEMDAN-GARCH model generalizes effectively beyond the training set. This approach minimizes the risk of overfitting and enhances the robustness of volatility predictions. The choice of R for this analysis allows for flexible and precise modeling, enabling a comprehensive investigation into the dynamic volatility patterns within aerospace and defense indices.

The sample

This study uses daily prices from aerospace and defense indices of four distinct countries, each represented by a specific index:

1. South Korea represented by the FR Korea Aerospace & Defense Price Return Index (TPXELDKPDAERO)

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0.0002 to 0.0003, indicating marginal average growth. Standard deviations reveal varying volatility, with China exhibiting the highest (0.02856) and the UK the lowest (0.0162), suggesting greater risk in Chinese aerospace and defense investments. Skewness values highlight asymmetric return distributions; China and South Korea display positive skewness, implying occasional large positive returns, whereas France shows negative skewness, indicating a higher occurrence of negative returns. The pronounced kurtosis across all indices, particularly in China (26.0278) and France (25.9313), points to leptokurtic distributions with heavy tails, reflecting the potential for extreme price movements. The Jarque-Bera test results, coupled with the highly significant p values (0.000***), strongly reject the null hypothesis of normality, confirming that returns are not normally distributed. These findings underscore the necessity for advanced volatility modeling techniques, such as CEEMDAN-GARCH, to accurately capture the complex dynamics and risks associated with these financial

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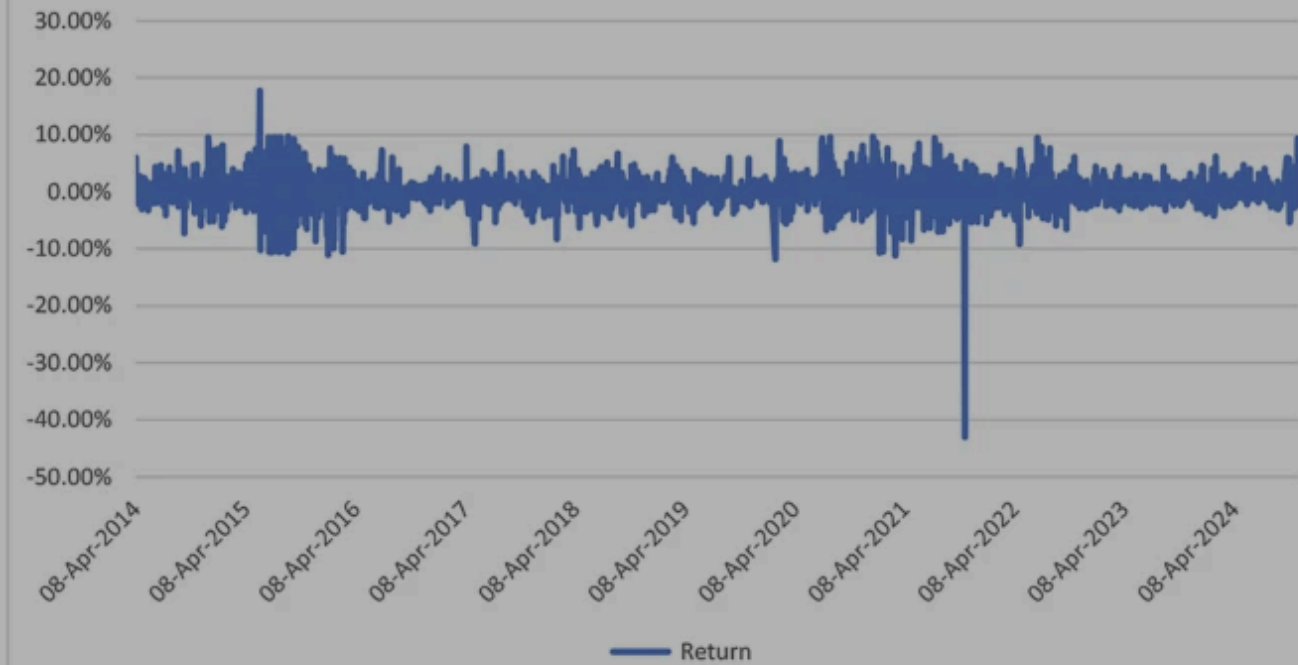
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China



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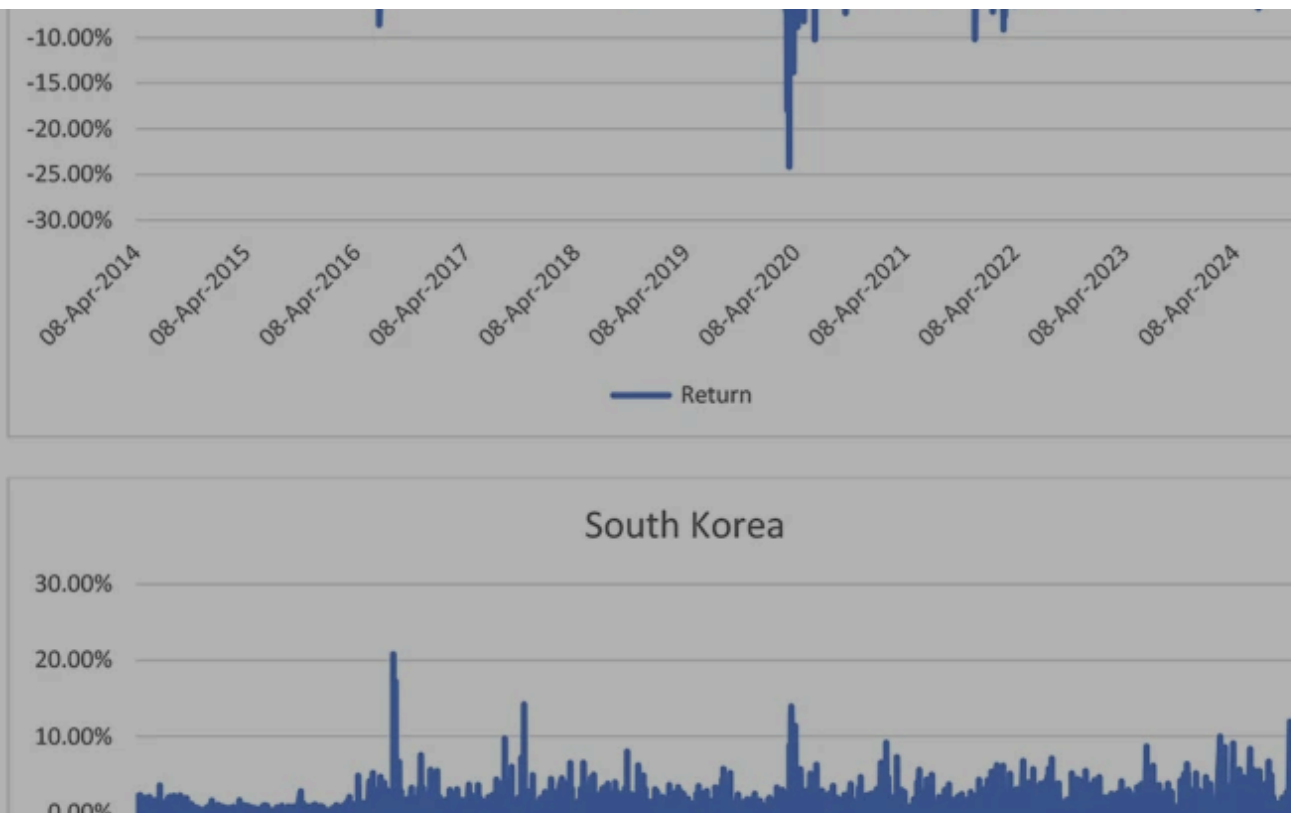
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Results and discussion

Our approach applied the CEEMDAN decomposition technique (in R studio) to the daily price data of the four selected aerospace and defense indices. Daily logarithmic returns were calculated for each index and subsequently processed using CEEMDAN to extract IMFs. These IMFs effectively isolate oscillatory components across different time scales, providing a detailed decomposition of the underlying dynamics (Fig. [2](#)).

Fig. 2

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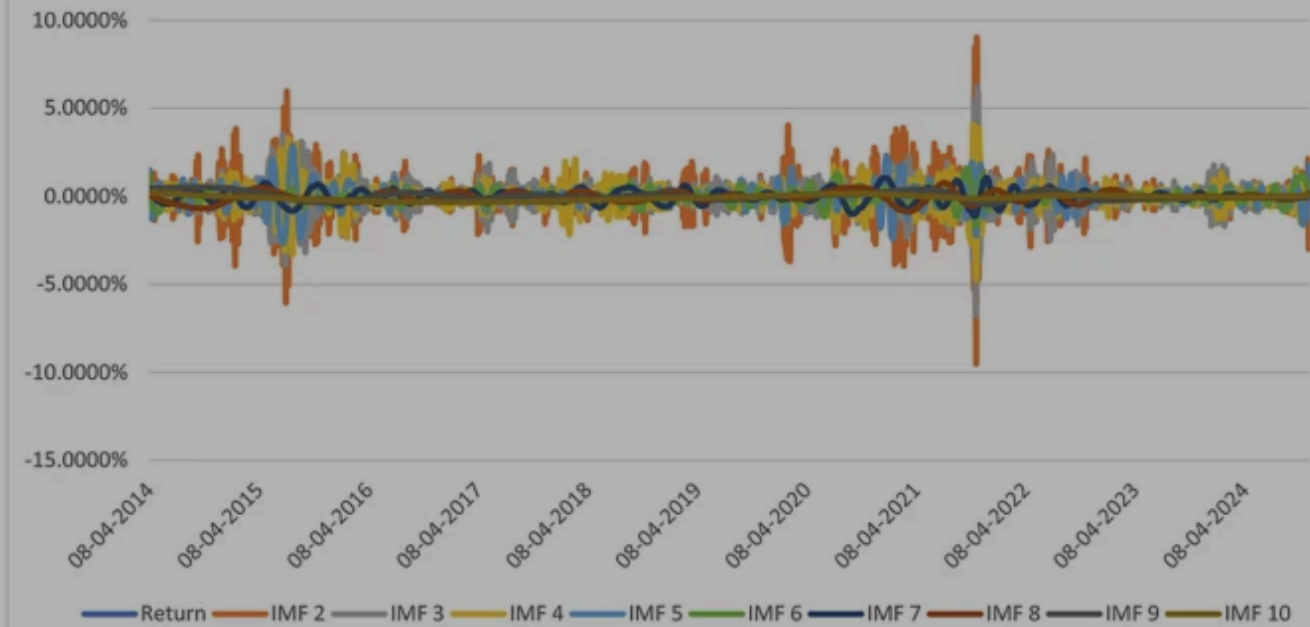
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China



South Korea

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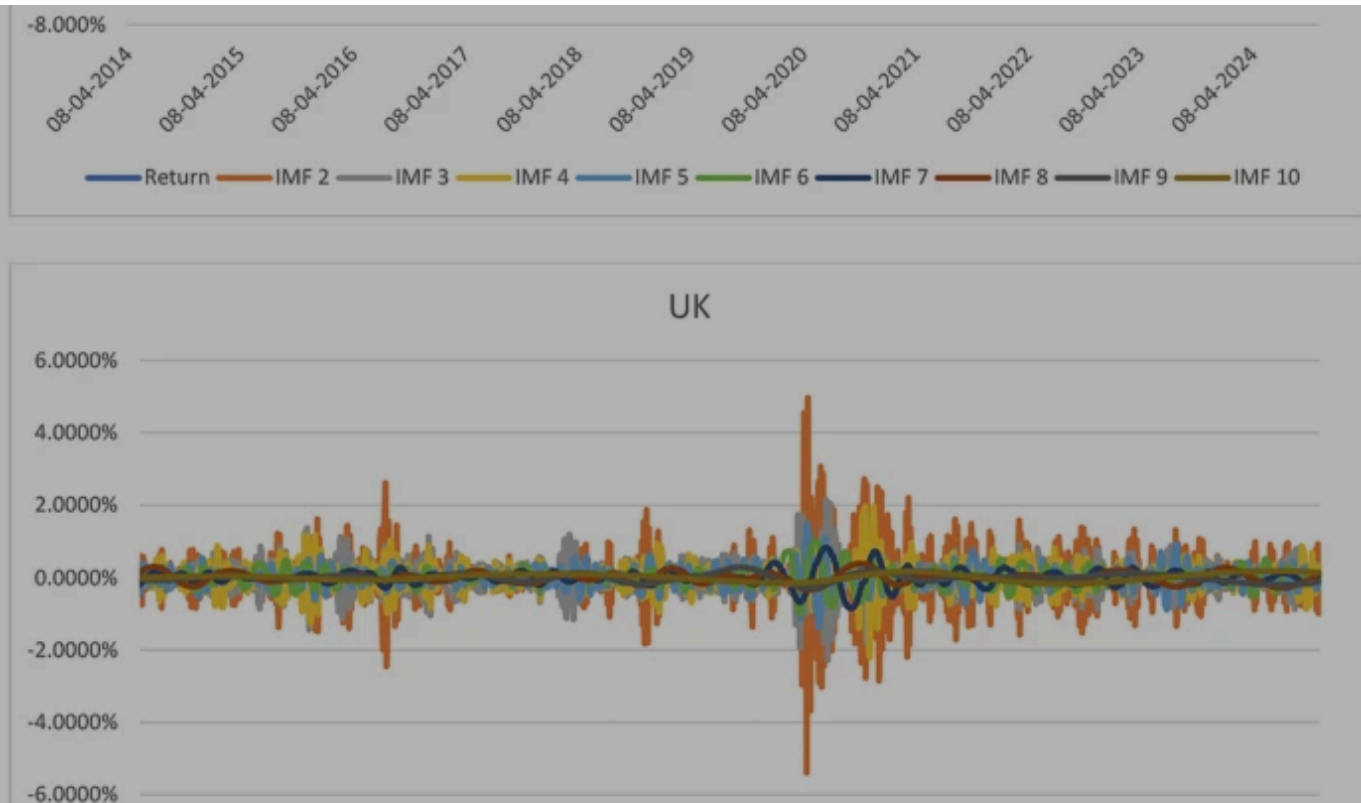
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asymmetries, allowing it to distinguish between heightened and subdued market conditions.

The in-sample period extends from May 15, 2014, to July 31, 2024, providing a comprehensive dataset to calibrate the GARCH models. To evaluate the predictive power of the proposed CEEMDAN-integrated framework, the parameters derived from the in-sample GARCH models were used to forecast volatility during the out-of-sample period, which spans 2 months from August 2, 2024, to November 8, 2024 (Fig. 3). This validation period allowed for a rigorous assessment of the models' performance, enabling a direct comparison between predicted and realized volatility values for each aerospace and defense index. For comparison and to isolate and quantify the impact of CEEMDAN in enhancing the predictive accuracy of volatility forecasts, the same GARCH models—IGARCH, SGARCH, and GJR GARCH,—were also applied directly to the raw

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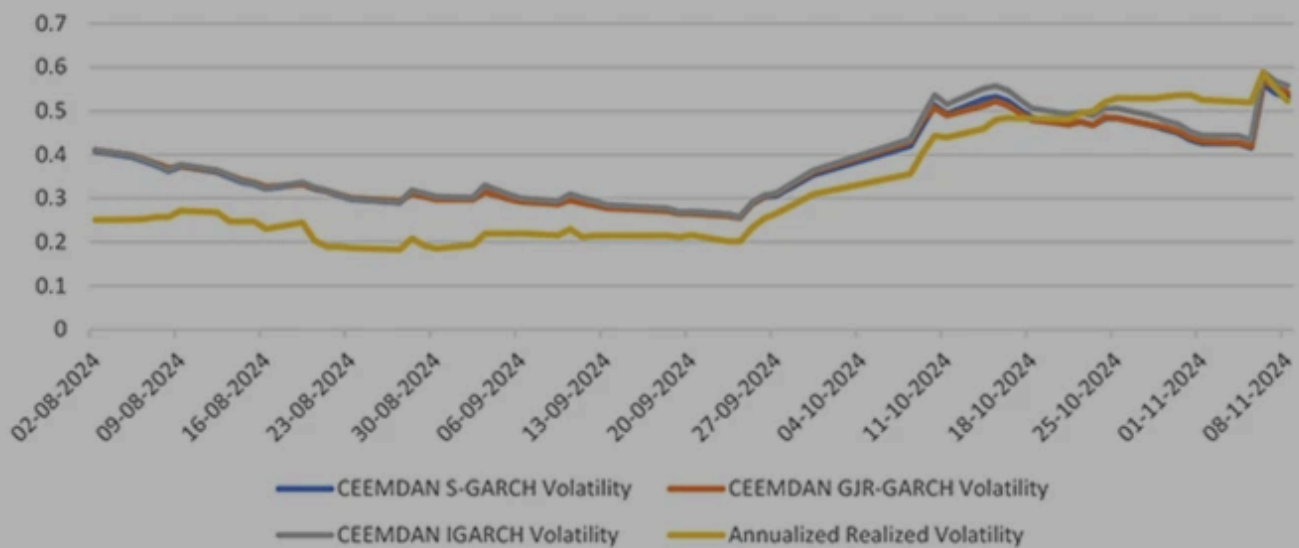
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China



South Korea

0.7

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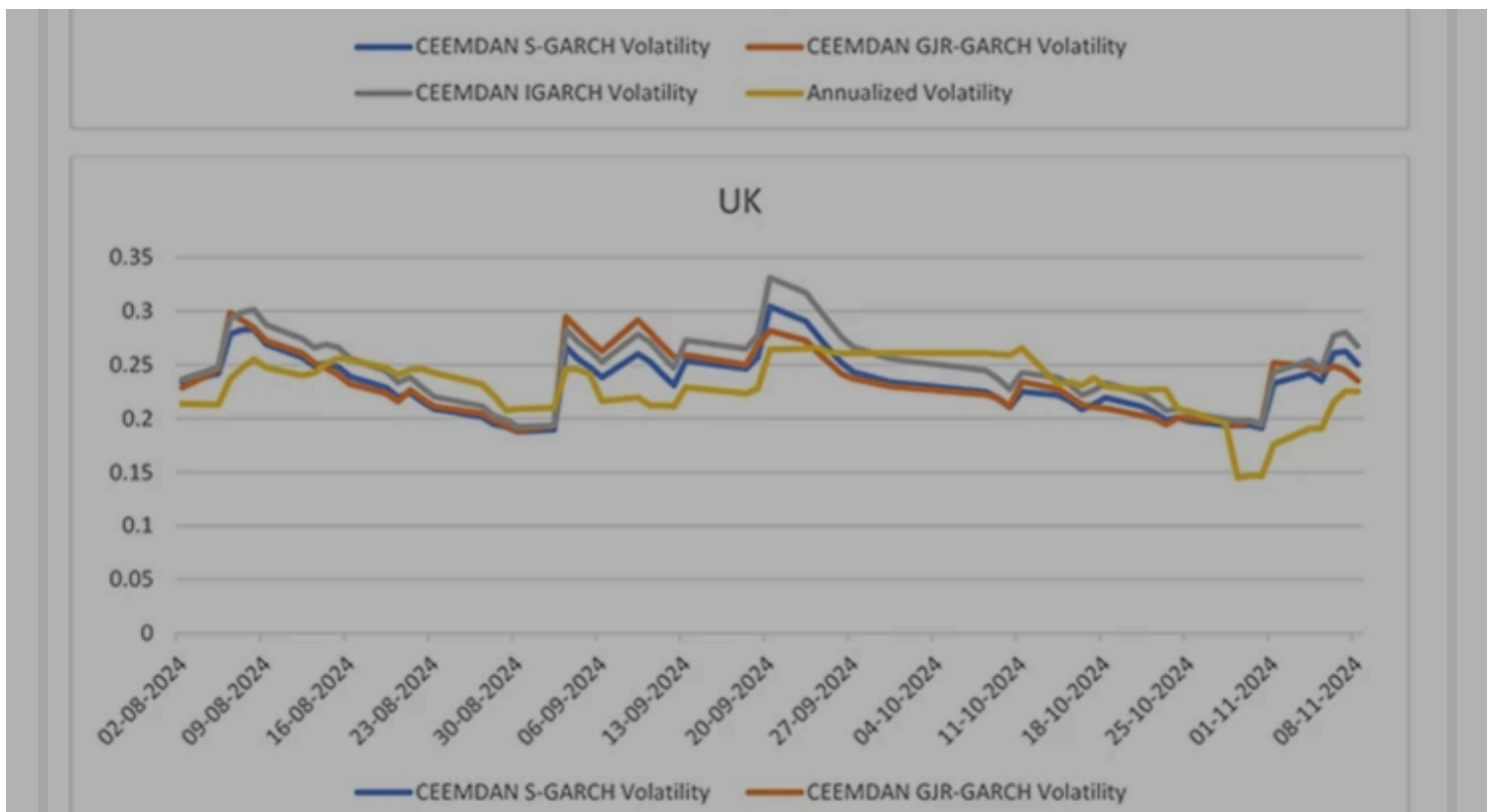
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parameter across all models, as evidenced by reduced p values (e.g., from 0.627 in SGARCH without CEEMDAN to 0.000049 with CEEMDAN). This indicates that the mean return is more precisely estimated after preprocessing. Additionally, the Akaike Information Criteria (AIC) values are slightly higher with CEEMDAN, suggesting a marginal trade-off in model simplicity for improved parameter estimation. CEEMDAN's influence on volatility-related parameters, such as α_1 and β_1 , remains consistent, with negligible changes in their significance and values.

For South Korea, the AR(1) parameter shows significant improvements in p values when CEEMDAN is applied to SGARCH and IGARCH, highlighting better capture of autocorrelations in returns. However, for GJR-GARCH, the p values remain high with or without CEEMDAN, indicating limited benefits in improving this model's fit for Korean data. Interestingly, the Log-Likelihood values slightly decrease after applying CEEMDAN,

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The Log-Likelihood values increase slightly for GJR-GARCH and IGARCH with CEEMDAN, indicating better explanatory power. Interestingly, AIC values improve (decrease) with CEEMDAN in these models, supporting the hypothesis that CEEMDAN enhances model performance for the UK market.

Results in Table 3 reveal key insights about the impact of CEEMDAN preprocessing on our model performance. Both in-sample and out-of-sample periods highlight nuanced changes across different models and indices.

For the FTSE China A 200 Aerospace and Defense Price Return Index, CEEMDAN shows limited improvement in model accuracy. In-sample MAPE for the best-performing model, GJR-GARCH, increases slightly from 14.67436 without CEEMDAN to 14.80969 with CEEMDAN. Similarly, out-of-sample results show negligible differences, with GJR-

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All-Share Aerospace and Defense Index, CEEMDAN preprocessing shows consistent improvements in out-of-sample accuracy. The best-performing model, SGARCH, improves from a MAPE of 11.73301 without CEEMDAN to 11.62022 with CEEMDAN. IGARCH and GJR-GARCH also show slight reductions in MAPE during the out-of-sample period, indicating enhanced prediction stability. However, in-sample results remain relatively stable, with IGARCH showing a negligible reduction in MAPE from 16.42273 to 16.30375.

Also, the analysis of the MAPE results shows consistent rankings of the best performing GARCH models across all indices. This indicates that while CEEMDAN enhances model precision in some cases, it does not fundamentally alter the hierarchy of model performance. For the FTSE China A 200 Aerospace and Defense Index, GJR-GARCH remains the best-performing model in both in-sample and out-of-sample periods, with

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Table 4 Statistical significance of out-of-sample MAPE differences with and without CEEMDAN.

The statistical tests indicate that CEEMDAN improves the predictive accuracy of IGARCH models, particularly for the China, France, and UK indices, where both the Wilcoxon signed-rank test and paired t -test yield highly significant p values ($p < 0.001$). This suggests that IGARCH, which is designed to capture long-memory effects in volatility, benefits from CEEMDAN's ability to refine time series decomposition by filtering out noise and enhancing signal clarity. Similarly, SGARCH models exhibit some improvements in forecasting performance, especially in France and the UK, further supporting the role of CEEMDAN in enhancing standard volatility models. However, in the case of South Korea, statistical significance is observed only in the paired t -test,

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those analyzed in this paper. Furthermore, (Zhang et al., [2015](#)) demonstrated that EEMD-GARCH models significantly enhance forecasting accuracy in volatile commodity markets, which parallels our findings that CEEMDAN improves the precision of GARCH models for the aerospace sector during geopolitical disruptions.

The observed variations in CEEMDAN's impact across the four indices can be closely linked to the unique geopolitical and economic contexts of the respective countries. In China, where the aerospace and defense sector has been a focal point of strategic state investment amid global trade tensions and technological competition, the relatively minimal improvement in forecasting accuracy may reflect the highly regulated and interventionist nature of Chinese markets, where external shocks and market forces are often moderated by government policies. Conversely, in France, the aerospace and defense sector is deeply integrated with European defense initiatives and export-driven

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Our results further emphasize the capacity of the CEEMDAN-GARCH framework to capture volatility dynamics influenced by major global disruptions, including the COVID-19 pandemic, the post-pandemic economic slowdown, and the ensuing inflationary pressures. The framework also reflects the impact of global monetary tightening policies, which have exacerbated market volatility, and supply chain disruptions that have disproportionately affected the aerospace sector. Additionally, financial shocks, such as the collapse of Silicon Valley Bank and broader debt crises, are mirrored in the indices' heightened volatility during specific periods. The geopolitical turmoil stemming from the Russia-Ukraine war not only increased demand for defense-related technologies but also introduced persistent uncertainty, which the CEEMDAN-GARCH framework effectively captured across all indices.

The robustness of the CEEMDAN-GARCH framework is reinforced by the results from

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emphasize CEEMDAN's potential to improve volatility forecasting, particularly for models sensitive to long-memory effects, while also highlighting its selective impact across different GARCH variations and market indices.

Conclusion

This study introduces a novel, exploratory framework for analyzing volatility in the aerospace and defense sector by integrating Complementary Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) into GARCH-type models (IGARCH, SGARCH, GJR-GARCH). Using daily index data from China, South Korea, France, and the United Kingdom spanning 2014–2024, we evaluated the forecasting performance of these hybrid models, particularly in the context of major global disruptions such as the COVID-19 pandemic and the Russia-Ukraine conflict.

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out-of-sample performance, in line with Tian and Tian ([2021](#)), who showed that the EMD-EGARCH model improves short-term prediction accuracy over standard EGARCH models. Similarly, Huang et al. ([2021](#)) highlighted the benefits of combining decomposition techniques with GARCH and deep learning models, while Zhang et al. ([2015](#)) found that EEMD-GARCH models significantly enhance forecasting accuracy in volatile commodity markets, mirroring our findings in the aerospace sector during geopolitical disruptions. This paper contributes to the niche literature on sector-specific forecasting by applying the framework to the underexplored aerospace and defense sector.

Contextually, the findings reinforce how market responses in this sector are shaped not just by financial fundamentals but also by broader security, policy, and ESG considerations. For instance, investor sentiment driven by defense spending shifts,

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applications should be approached with caution, given the modest scale of forecasting improvements observed.

Future research can integrate machine learning techniques with CEEMDAN to enhance volatility forecasting. Deep learning models like LSTM, GRU, and Transformers (e.g., Temporal Fusion Transformer) can improve prediction accuracy, while XGBoost and Random Forest can aid in IMF selection. Bayesian optimization can further refine GARCH model selection. Beyond aerospace and defense, this framework could be applied to cryptocurrencies, renewable energy, and technology markets, where volatility is high. Additionally, its use in emerging markets could improve risk assessment. A comparative analysis of CEEMDAN vs. Variational Mode Decomposition (VMD) could also assess the best decomposition approach for financial time series forecasting.

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The data utilized in this study consists of daily price data for aerospace and defense indices, sourced from the Refinitiv database. These datasets are not publicly available but can be accessed through Refinitiv with the appropriate institutional subscription. The data can be obtained from Refinitiv upon request or from the corresponding author on reasonable request.

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Author information

Authors and Affiliations

Notre Dame University—Louaize, Zouk Mosbeh, Lebanon

Viviane Naimy, Tatiana Abou Chedid, Omar Abou Saleh & Nicolas Bitar

Contributions

VN led the study design, oversaw the research process, and provided detailed interpretation of the results. NB and TAC contributed to the drafting and writing of the

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